

AI-Powered Educational Admission Assistant

Retrieval-Augmented Generation (RAG) Based Intelligent Admission Support System

IBM SkillsBuild Internship Project

Intern: Gugulothu Bhavith

AICTE Student ID: STU696c4be1c29041768704993

Institute: JNTUH University College of Engineering Palair

Duration: 15 May 2026 – 15 June 2026

Problem Statement

Background: Educational institutions receive a large number of admission-related inquiries from prospective students, parents, and guardians throughout the admission cycle. These inquiries typically involve eligibility criteria, admission procedures, fee structures, scholarship opportunities, reservation policies, academic regulations, required documentation, and available academic programs. Managing such requests manually often results in delayed responses, inconsistent information delivery, and increased workload for administrative staff.

Problem Identification: Most educational institutions lack an intelligent and automated system capable of understanding natural language queries and retrieving accurate information directly from official admission documents. Students frequently struggle to obtain timely and reliable information, while admission departments face challenges in handling repetitive queries efficiently. Traditional support mechanisms are often time-consuming, resource-intensive, and difficult to scale during peak admission periods.

Challenges: The primary challenges include processing large volumes of admission documents, retrieving relevant information accurately, understanding diverse user queries, minimizing AI hallucinations, maintaining document-grounded responses, ensuring scalability for thousands of users, and delivering personalized admission guidance with high consistency and reliability.

Proposed Solution: To address these challenges, an **AI-Powered Educational Admission Assistant** has been developed using **IBM watsonx.ai**, **IBM Granite Foundation Models**, **Langflow**, **ChromaDB Vector Database**, **NVIDIA BAAI BGE-M3 Embeddings**, and **Retrieval-Augmented Generation (RAG)** architecture. The system ingests institutional documents, converts them into semantic vector embeddings, stores them within a vector database, retrieves relevant information based on user queries, and generates context-aware responses grounded in official institutional knowledge.

Expected Impact: The proposed solution aims to improve accessibility to admission information, reduce administrative workload, provide 24/7 intelligent support, enhance transparency in the admission process, improve response accuracy, and deliver a modern AI-driven experience for prospective students and parents. The system demonstrates the practical application of enterprise-grade artificial intelligence in transforming educational admission support services.

Keywords: *Artificial Intelligence, IBM watsonx.ai, IBM Granite Foundation Models, Langflow, ChromaDB, NVIDIA Embeddings, Retrieval-Augmented Generation, Educational Technology, Admission Assistant, Vector Database, Natural Language Processing*